"The Spark Plugs of Life"

Mineral Resources

Minerals

A mineral is a naturally occurring substance that is solid and stable at room temperature, representable by a chemical formula, usually abiogenic/inorganic, and has an ordered atomic structure.



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http://www.ima-mineralogy.org/Medal.htm

As of July, 2015, there are 5048 valid species of minerals in the list of International Mineralogical Association (IMA). The silicate minerals compose over 90% of the Earth's crust. Silicon and oxygen constitute approximately 75% of the Earth's crust, which translates directly into the predominance of silicate minerals.

Minerals can be described by various physical properties which relate to their chemical structure and composition.

Common Distinguishing Characteristics:

Crystal structure and habit, hardness, luster, diaphaneity, color, streak, tenacity, cleavage, fracture, parting, and specific gravity. More specific tests for minerals include reaction to acid, magnetism, taste or smell, and radioactivity.

Five (5) Requirements a Material has to meet to be considered as Mineral:

a. Naturally occurring
   - Means that a substance is not man-made
   - Exists by nature without artificial aid

b. Inorganic/Abiogenic
   - *Inorganic* means a substance without a carbon basis and is synthesized through geological systems and processes.
   - *Abiogenic* means not produced by living organisms.

c. Stable at room temperature
   - It has to stay solid in room temperature (20-25°C).
   - Simply, a mineral is a solid.

d. With a definite chemical composition

   As defined by Hobart King in the article "What are Minerals", "Definite chemical composition" means that all occurrences of that mineral have a chemical composition that varies within a specific limited range.

   For example: the mineral halite (known as "rock salt" when it is mined) has a chemical composition of NaCl. It is made up of an equal number of atoms of sodium and chlorine.

e. Has an ordered internal structure

   This means that the atoms of a mineral must be arranged in a systematic and repeating pattern.

Mohs Hardness Scale

	Mineral	*Chemical formula*	*Absolute Hardness*
1	Talc	$Mg_3Si_4O_{10}(OH)_2$	1
2	Gypsum	$CaSO_4 \cdot 2H_2O$	3
3	Calcite	$CaCO_3$	9
4	Fluorite	CaF_2	21
5	Apatite	$Ca_5(PO_4)_3(OH–, Cl–, F–)$	48
6	Orthoclase Feldspar	$KAlSi_3O_8$	72
7	Quartz	SiO_2	100
8	Topaz	$Al_2SiO_4(OH–,F–)_2$	200
9	Corundum	Al_2O_3	400
10	Diamond	C	1600

Talc, the number 1 on the hardness scale, is the softest and can be easily scratched; diamond, the last on the hardness scale, is the hardest among all minerals.

The Mohs scale

- measures the hardness of a material
  The hardness of a material is measured against the scale by finding the hardest material that the given material can scratch, and/or the softest material that can scratch the given material. For example, if some material is scratched by apatite but not by fluorite, its hardness on the Mohs scale would fall between 4 and 5
- is a purely ordinal scale
  Example: Corundum (9) is twice as hard as topaz (8), but diamond (10) is four (4) times as hard as corundum.

Minerals as Indicators of Specific Environment Conditions

Minerals can be used as indicators of specific environmental parameters on a variety of scales ranging from the micro (thin section) to macro (hand specimen) to profile (vertical differences) to landscape (hydrology, topography) to even continental scale.

The Environment Conditions

pH – Acid or Alkaline with dominant anion

The pH is the master variable and pH of surface and ground waters (shallow and deep) have a direct effect on the precipitation and stability or persistence of secondary minerals in the regolith (a product of weathering)

Redox Conditions

Minerals form and persist in a restricted pH range of oxidation potential. Pyrite (black) and mono sulphides ("black ooze") are indicators of reducing environments, where sulphur reducing bacteria catalyze the oxidation of organic matter and link it to the reduction of Fe^{3+}.

Drainage or water activity

The rate of flow of groundwater vertically and laterally (landscape) through a profile, sediments or sedimentary rock can affect the nature of the minerals that form, and the presence of resulting minerals can be used as indicators of the rate of flow of water or in thermodynamic terms the water activity.

*Property of STI

Landscape or Catenaries' Position

Regolith minerals can be used as indicators of past landscape position, but the interpretation is a combination of drainage and pale landscape evolution or substrate type. The iron oxides, especially hematite and goethite, having their genetic pathways controlled significantly by water activity or saturation, provide the best indicators of drainage linked to the centenary position.

Temperature & Climate

Generally, there are few reliable mineral indicators of current and past temperatures. The main indicators are the iron oxides and hydroxides. Magnetite, commonly found in globules (nodules, mottles) within the soil and on the surface as ferruginous lag, is a likely indicator of forest-fires. Iron oxides (goethite, hematite) present in soil matrix and in globules when heated to temperatures above 3000℃ in the presence of organic matter (top soil organics), transform to magnetite. Therefore, the presence of magnetite is used as an indicator of forest fires.

Transported vs. in situ regolith

Regolith is broadly classified into that produced as a result of weathering of the basement rocks in situ and that produced due to the transportation processes, which is essence is surficial sediments.

Importance of Minerals

- Presence of minerals in the food we ingest keeps us energized and active
- Construction industry
  Minerals are used in building of foundations, road bases, concrete, and drainage, require crushed stones. Sand and gravel are also utilized for the construction of concrete and foundations. Clays are utilized to make cement, bricks and tile. Iron ore is consumed for reinforcing rods, steel beams, nails and wire. Dimension stone is important for architectural work as they are used for facing, curbing, flooring, stair treads.
- Agriculture
  Fertilizers are made using phosphate rock and potash. For acid-neutralizing soil treatment, Lime is used. Mineral nutrients are also added to animal feeds to keep them healthy.
- Chemical industry
  Salt, lime and soda ash are used widely in the chemical industry. Amounts of metals, clay and mineral fillers/extenders are used in manufacturing.

Furthermore, typical examples of industrial rocks and minerals are limestone, clays, sand, gravel, diatomite, kaolin, bentonite, silica, barite, gypsum, and talc. Some examples of applications for industrial minerals ceramics, paints, electronics, filtration, plastics, glass, detergents, and paper.

The evaluation of raw materials to determine their suitability for use as industrial minerals requires technical testwork.

Mineral Development and Environmental Damage

Various entities such as air, water, land, vegetation, animals including human, landscape and geomorphologic features, historical heritage, etc. are adversely affected one way or the other during the course of mineral development.

Air, land, and water pollution

Varying degrees of pollution of air, water and land occur in the course of mineral development depending on the stage and scale of activities attained. While only minor pollution occurs during mineral exploration, more intense air and water pollution emanates from the exploitation stages, particularly if carried out on a large scale. In the oil-producing areas of the country oil spillage of differing intensity resulting from burst pipelines, tanks, tankers, drilling operations, etc. is a common phenomenon.

Damage of vegetation

Vegetation in the form of natural forest or crop plantation is usually the first casualty to suffer total or partial destruction or degradation during the exploration and exploitation of minerals in a locality. The vegetation damage is more extensive at the time of mine development and mining operations and is more expensive when crop plantation is affected.

Ecological disturbance

This may not be immediately felt but mineral extraction and processing activities cause disturbance of the ecosystem with possible adverse consequences on the floral and faunal community in general. Oil spillage produces a devastating ecological disturbance in the oil-producing states as well as in areas where leakage occur due to natural breakage of oil pipe line or illegal bunkering .The plants, animals, soils, and water are affected.

Degradation of natural landscape

A common negative effect of mining minerals is creating open space in the ground and generating heaps of rock wastes that cannot be easily disposed off.

Geological hazards



Blasting
Retrieved from
safedistanceforblasting.wikispaces.com

Mining operations normally upset the equilibrium in the geological environment, which may trigger off certain geological hazards such as landslide, subsidence, flooding, erosion and tremors together with their secondary effects. Minor earth tremors are generated due to blasting of rocks in various quarries (an excavation or pit, usually open to the air, from which building stone, slate, or the like, is obtained by cutting, blasting, etc.) Villages and settlements in the neighborhood of the quarries have experienced unpleasant earth movements when the rocks are blasted.

Minerals and Rocks

Minerals are naturally occurring inorganic crystalline substances with physical and chemical properties with prescribed limits. Rocks are aggregates of a mineral or minerals. There are over 5000 minerals, however only a few are necessary to identify most of the rocks.

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